

OpenLoop

The AI case agent for bank files that stay open for weeks

Code enforces the rules · Claude plans the next step · people make the decisions

NEXXT AI Hackathon 2026 · Raiffeisen Bank Romania · AI for Operational Efficiency · Petru & Teodor

Back-office work is an open loop

Documents arrive over days; every one must agree with the others and with the policy

95 days

and about $2,598 per corporate KYC review (2023 vendor survey)

30 days

a trade-register certificate stays valid, then the file must ask for a new one

14 days

to report a beneficial-owner mismatch (AMLR art. 24, from July 2027)

Today a person chases every missing paper, **re-checks** every expired certificate and **reconciles** owners and addresses by hand.

“Procesarea desfăşurându-se manual” — Raiffeisen Bank Romania on a 14,000-client remediation (research/01)

The risk is in the sequence, not the document

Two real data-protection fines against Raiffeisen Bank Romania

2022 Credit-bureau queries without the applicant's signed consent; a client's phone number replaced with an employee's

2024 An employee changed clients' phone and email details, then applied for credit products — all without consent

OpenLoop watches the ledger: SR-01 flags a staff contact change followed by a credit action within 48 h with no client confirmation. SR-02: the client confirms or disputes the change in the portal; a dispute opens a card that can put the file on hold.

OpenLoop in one picture

One append-only, hash-chained ledger per file — every actor reads it and writes to it

Events in

documents · emails · portal uploads · staff actions · time passing

→

Code pipeline

extraction with verbatim quotes · registry lookup · hard rules HR-01…08 · timers and expiry

→

Claude plans

Opus 5.5 on Amazon Bedrock decides the next step over judgment-only tools

→

Gate

only actions the rules allow can run · approve credit is always DENIED

→

People decide

short decision cards · rulings become precedents and learned checks

Judge Featherless classifier + similarity check — auto-accept only when both agree

Two screens, one ledger the bank's work queue and the client portal, each in its own words

One file, start to decision

The bakery's SME loan and KYB file — the demo, in six moments

1

Day 0

4 documents arrive; DSCR 1.4 verified; the agent asks for what's missing

2

Register

an undeclared 25% owner starts the legal clock; the address is written two ways

3

The gate

a client email arrives; the agent tries to ask for a new declaration — denied until a person decides

4

Decided

two analyst clicks; corrected documents; the clock is cancelled; file complete

5

Expiry

the certificate expires a week later; the engine reopens the file by itself

6

Insider

a staff phone change, then a credit check; the client says “No, I didn't”; file on hold

The credit decision stays human: the agent has no tool that approves credit.

Agent design: a hybrid by evidence

Code where rules are certain, Claude where judgment is needed, people where it matters

Code always runs

extraction with verified quotes, registry, rules, timers, expiry — deterministic

Claude plans

next steps, email intent, escalation wording; 20 s budget per click

Fails safe

errors or timeouts fall back to the scripted planner; a code safety net runs after Claude

Gate on every action

allowed by the rules or denied and logged (GATE-01/02/03)

Humans in the loop

cards show only what differs; rulings become precedents

Learns safely

3 similar rulings propose a check; backtested; a different person approves

Measured live: 3.5–13 s per click, 0 fallbacks · every step logged as pipeline / claude / planner / safety-net

Tested, calibrated, measured

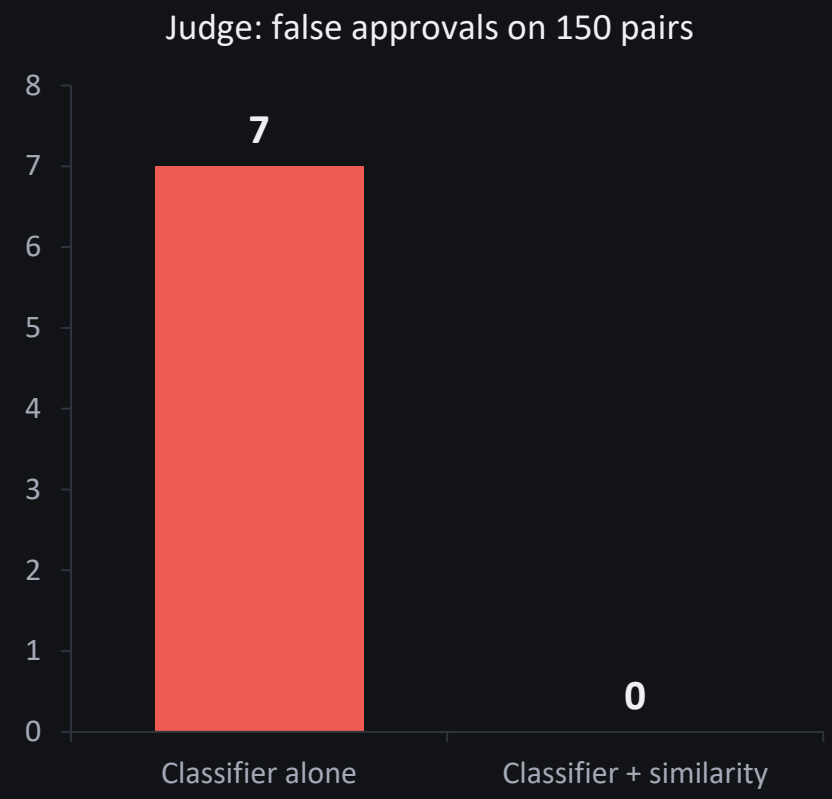
Evidence in the repo, reproducible offline

132

automated tests: the full demo, every rule path, role leaks, tamper detection

8 / 8

synthetic companies end on their intended problem (one rule path each)



8 → 0

false approvals of the learned address check after small-town counterexamples

390–2560

pixel widths browser-tested, both apps, no console errors

Reproducible in one command

FastAPI · SQLite event store · React bank app and client portal · Claude on Bedrock · Featherless

```
pip install -r requirements.txt
cd web && npm ci && npm run build && cd ..
python -m uvicorn app.main:app --app-dir backend
# bank /   ·   client portal /client/
```

Offline mode (scripted planner, heuristic judge) runs the whole demo with no API keys. **Live mode** swaps in Claude and the Featherless judge with the same UI and the same ledger.

- **Data stays in the bank**

  one append-only table; PostgreSQL on-prem in production

- **Bring your own document AI**

  extractors are adapters; quotes are verified by code

- **Privacy by construction**

  client views are allow-lists; staff identities never reach clients

- **All synthetic data**

  documents follow real ONRC / ANAF / F20 forms; tax IDs in an unassigned range

Romanian rules, in code

Two templates on one engine: SME loan and KYB check · Annual business review

HR-01	CUI checksum
HR-02	Company active in the trade register
HR-03	Certificate under 30 days old
HR-04	Owners ≥25% match the register (Law 129/2019)
HR-05	An ID for every owner
HR-06	Same registered office everywhere
HR-07	Debt service coverage ≥ 1.2
HR-08	No tax debts on the ANAF certificate
SR-01	Staff contact change → credit action in 48 h
SR-02	Client disputes a staff change

Judge questions JQ-01 (same address?) and JQ-02 (signatory = administrator?) decide only after the hard facts match.

Where it pays off

The agent does the work between human decisions

100,000+

active Smart Business SME clients at Raiffeisen Romania (H1 2026)

1 or 5 yrs

AMLR periodic refresh by risk, from July 2027: every SME, again and again

Path to production

1 **Shadow mode**

annual business reviews with one back-office team

2 **SME onboarding**

the full loan and KYB template

3 **Same engine**

garnishments (2.25M ANAF acts in H1 2025), DORA incident clocks, complaint deadlines

Analysts see only files that need a decision · clean files finish on their own · every action is auditable

Why it's new

Time is first-class

the file is a ledger; rules run over sequences, not single documents

An agent that cannot overstep

a gate between plans and actions; credit decisions stay human

It learns from people, safely

rulings → precedents → backtested checks with maker-checker approval

One ledger, two audiences

the bank and the client see the same file, each in their own words

A 4-week shadow pilot

on annual business reviews with one back-office team

- **Plug in** the bank's document AI and the ONRC / ANAF data feeds, on-prem
- **Measure** hours per file, days to complete, re-requests and audit findings
- **Decide** with numbers from the bank's own files

OpenLoop · github.com/Petsteb/openloop-case-engine (private) · synthetic data only